QUASI-COMPACT SEPARATED LOCALLY QUASI-FINITE MORPHISMS FROM A SCHEME GIVE A REASONABLE SPACE

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Bounded fibres over quasi-compact source. Morphisms, lemma locally-quasi-finite-qc-source-universally-bounded (cross-chapter, named). The bound n on the fibres of U -> X.
- Permanence under etale surjection. lemma-universally-bounded-permanence (same chapter, carded L26046). Transfers the bound to all etale morphisms into X.

Source reference. Original lemma 086T: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1715–1715. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAQ: focused-seeds/geometry/S9-decent-get-reasonable/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03FZ: focused-seeds/geometry/S9-decent-get-reasonable/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Assume*

(1) *every quasi-compact scheme étale over X has finitely many irreducible components, and*
(2) *every $x \in |X|$ of codimension 0 on X can be represented by a monomorphism $\mathrm{Spec}(k) \to X$.*

Then X is a reasonable algebraic space.

Proof. Let U be an affine scheme and let $a : U \to X$ be an étale morphism. We have to show that the fibres of a are universally bounded. By assumption (1) the scheme U has finitely many irreducible components. Let $u_1, \ldots, u_n \in U$ be the generic points of these irreducible components. Let $\{x_1, \ldots, x_m\} \subset |X|$ be the image of $\{u_1, \ldots, u_n\}$. Each x_j is a point of codimension 0. By assumption (2)

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we may choose a monomorphism $\mathrm{Spec}(k_j) \to X$ representing x_j. By Properties of Spaces, Lemma 0BAQ we have

$$U \times_X \mathrm{Spec}(k_j) = \coprod\nolimits_{a(u_i)=x_j} \mathrm{Spec}(\kappa(u_i))$$

This is a scheme finite over $\mathrm{Spec}(k_j)$ of degree $d_j = \sum_{a(u_i)=x_j} [\kappa(u_i) : k_j]$. Set $n = \max d_j$.

Observe that a is separated (Properties of Spaces, Lemma 03FZ). Consider the stratification

$$X = X_0 \supset X_1 \supset X_2 \supset \ldots$$

associated to $U \to X$ in Lemma 086T. By our choice of n above we conclude that X_{n+1} is empty. Namely, if not, then $a^{-1}(X_{n+1})$ is a nonempty open of U and hence would contain one of the x_i. This would mean that X_{n+1} contains $x_j = a(u_i)$ which is impossible. Hence we see that the fibres of $U \to X$ are universally bounded (in fact by the integer n). $\square$